

Paris Road Traffic Conditions 2023

1.Dataset Overview

- **t_1h (Timestamp):** Records the exact time of the observation. The data is aggregated at a 1-hour temporal resolution.
- **iu_ac (Segment ID):** A unique identifier for each specific detector within the Parisian road network (e.g., 6409, 6414). The dataset encompasses a total of 2,982 physical monitoring sites.
- **etat_traffic (Traffic State):** Indicates the current congestion level of the road segment. It is a categorical target variable with five distinct states: Inconnu (Unknown), Fluide (Free-flowing), Pré-saturé (Dense/Pre-saturated), Saturé (Saturated), and Bloqué (Blocked/Gridlock).
- **q (Traffic Flow):** Represents the traffic volume, defined as the total number of vehicles passing through the detector's cross-section per unit of time (vehicles/hour).
- **k (Average Speed):** Denotes the average spatial speed of vehicles traveling along the given road segment (measured in km/h).
- **lon & lat (Spatial Coordinates):** The geographic longitude and latitude of the road segment, essential for geospatial mapping and network topology analysis.

```
library(arrow)
library(dplyr)
library(knitr)

file_path <- "paris_traffic_part_000.parquet"
traffic_df <- read_parquet(file_path)

traffic_df %>%
  head(5) %>%
  kable(caption = "Paris Road Traffic")
```

Table 1: Paris Road Traffic

t_1h	iu_ac	etat_traffic	q	k	lon	lat
2023-01-01 03:00:00	6409	Inconnu	NA	NA	2.381947	48.86808
2023-01-01 03:00:00	6410	Inconnu	NA	NA	2.381638	48.86800
2023-01-01 03:00:00	6412	Inconnu	356	NA	2.379740	48.86945
2023-01-01 03:00:00	6413	Inconnu	NA	NA	2.378849	48.87047
2023-01-01 03:00:00	6414	Fluide	356	6.67333	2.378589	48.87035

2.Problem

Among 250,000 data rows, q has only 116,092 non-missing values (a missing rate as high as 53.5%), while k has a missing rate of approximately 52.6%. Additionally, in the target label distribution, Inconnu (unknown status) accounts for 131,605 entries, representing about 52.6% of the total.